

DI22001 - Algorithms and Artificial Intelligence

AI Project Assignment

This assignment is designed to give you an opportunity to apply Artificial Intelligence to a real-world problem, by defining a problem, obtaining relevant data, modelling it as needed according to the definition of the problem, obtaining insights using the search strategies covered in the module, and linking your work to relevant aspects of algorithms and their complexity. This assignment is to be conducted **in teams of 5 students** and is worth **30% of the module mark**.

Brief

Building on lecture materials and lab exercises, your task is to identify an interesting problem that can be solved, at least in part, by **three of the search strategies we have studied** in the AI part of the module. The choice of problem is unrestricted, but the size of the work must be suitable for the available time (almost 6 weeks) and group size (5 people).

In choosing your problem and planning your work, keep in mind the four questions we have discussed in class:

1. What is the AI problem that we want to solve?
2. Why it is an important problem to solve?
3. How can we solve it?
4. How can we solve it in a better way (faster, more robust, and efficient)?

You don't need to produce a project proposal this year, but you are still encouraged to **discuss your ideas** with us. Be prepared to describe your problem and its significance, the dataset you will use, and what results you expect to get from the analysis. We will provide feedback in terms of scope and suitability of the project. Based on this feedback, you should **carry out any preparations of the data and** infrastructure to solve your problem. You should **collect and interpret any results** in terms of the problem definition, of the performance of the chosen search strategies, and of their real-world meaning. Finally, you should **critically assess your work**, identifying challenges, limitations, and opportunities for improvement. Pay special attention to aspects of your approach that link the analysis to the algorithms involved and their complexity (e.g. if you are forced to reduce the size of your dataset so that you can run the analysis algorithms within your time and memory constraints).

You will report on your work in a **15-minute presentation** to the lecturing team and the rest of the class. The presentation should focus on the problem and its relevance, and the results and interpretation of your results, avoiding technical details. At the end of the project, you will submit a written report in which you can describe in more detail your approach, the strategies you used, your results and their interpretation, possible future enhancements, and any relevant considerations linking your work to relevant algorithms and their complexity.

Above and beyond

Consider going beyond the project brief, by incorporating advanced techniques that were briefly covered in class. Possible extensions include, but are not limited to:

- using CSP, planning to solve part of your problem, in addition to the three search strategies
- developing a good, accessible interface to allow users to interact with your tools and data
- using machine/deep learning techniques to solve part of your problem

Up to 20% of the project's mark will come from extensions to material covered in class.

Submission & Assessment

This coursework is due for submission via MyDundee (under Assessment Area > Assessment: AI Project > ...), including presentation slides, final report, and implementation code.

Deadlines:

- The project assignment is due on **Wednesday 10 June** (Week 15).

Project presentations will take place during Week 15 (presentation schedule to be announced during Week 14). The **slides and presentation are worth 30%** of your total grade for this assignment. The **report is also worth 30%** of your total grade for this assignment. The **implementation code is worth 20%** of your total grade for this assignment. The **remaining 20%** of your total grade for this assignment will be given based on **above and beyond** as explained above.

Submissions should be made by one member of your team. All elements submitted should include the names and matriculation numbers of all team members.

Your submissions should be as follows:

- **Professional-looking slides** for a 10-minute presentation as a PowerPoint or PDF document, including the title of your project, the names of the participants, a description of the problem and its relevance, a description of your dataset and your approach to solving the problem using search strategies, the results of your analysis and their interpretation, anything you have done that is above and beyond the scope of the brief, and a conclusion; **All the implementation code** that you created and used in your assignment as a ZIP file, readable, fully commented and with suitable references to any code re-used; **A 2-page professional-looking report** (~1500 words, excluding any references and figure) as a Word or PDF document, with word count given, describing your

problem, the dataset, your approach to solving it, the results you obtained and their interpretation, possible future enhancements, and any relevant considerations linking your work to the algorithms involved and their complexity.

Check that the contents of your files are correct; keep a copy of what you submit in case there are problems with your submission.

Peer Reviews

The marks and feedback resulting from your submissions will be your group's mark. Individual marks are calculated based on the peer reviews, and this is the only way we can incorporate individual contributions to groupwork. It is essential that **all team members submit a peer review** which quantifies their own and other members' contributions to the work.

Peer reviews are moderated by the module lecturers and any changes to an individual mark (upwards or downwards) from the team mark will result from consensus in peer reviews from the entire team as to how much an individual contributed to the total workload.

If you have awarded someone in your team a greater or lesser contribution than others, please provide comments to justify your decision. In doing so, you may wish to reflect upon the following:

- were tasks outlined and were these agreed and clear to everyone in the group and when the deadlines were;
- did people contribute and complete their tasks to the best of their ability;
- where tasks completed on time;
- did people attend group meetings;
- did people communicate well and keep each other aware of progress;
- were the contributions adequate and to an acceptable standard (bearing in mind different skill levels within the team).

Peer reviews will be submitted via MyDundee. An assignment will be set up nearer the time of the deadline.

Presentations

Each team will be allocated a 15-minute slot for presenting their work. Presentations should last **10 minutes**, with 5 minutes left for questions from the audience. **Every team member should be present in the presentation, but not everyone must speak.**

Avoid technical details in your presentation and focus on the 4 questions above. Describe your problem and motivate its relevance. Present the source and properties of your data. Discuss the search strategies and, if applicable, any other techniques you used to solve the problem. Describe the results and, crucially, their interpretation in terms of the real-world problem you chose to address. Motivate your audience to engage with the problem and with your results.